Appendix 5. Results on amount and intensity of physical activity.

Arthroplasty/Measurement method/Study	Age/BMI/gender	Follow-up time	Outcome
Total hip arthroplasty			
Objective meaurement devices			
Alvarez et al. 2015	N: 47 Age: 63.8±11.7 55% male BMI: 29.6±5.6	>1 year	148.9±69.8
Clement et al. 2019	N: 200 Age: 69.9±9.2 (42-92) 43% male	12 & 24 months	*Patients aged <65:* Steps: 12 mo: 5370±2524; 24 mo: 4269±2905 steps/day Energy expenditure: 12 mo: 31.1±1.4; 24 mo: 30.4±1.2 MET/day Time spent walking: 12 mo: 1.2±0.5; 24 mo: 1.0±0.6 hrs/day Time spent standing: 12 mo: 3.4±1.6; 24 mo: 3.0±1.3 hrs/day Time spent sitting/lying down: 12 mo: 18.3±2.2; 24 mo: 18.6±2.0 hrs/day *Patients aged 65-74* Steps: 12 mo: 6071±3256; 24 mo: 5760±3199 steps/day Energy expenditure: 12 mo: 32.2±1.7; 24 mo: 31.1±1.5 MET/day Time spent walking: 12 mo: 1.3±0.7; 24 mo: 1.2±0.6 hrs/day Time spent standing: 12 mo: 4.4±2.0; 24 mo: 4.3±2.2 hrs/day Time spent sitting/lying down: 12 mo: 17.6±2.9; 24 mo: 17.6±2.7 hrs/day *Patients aged ≥75* Steps: 12 mo: 4344±2195; 24 mo: 3915±2054 steps/day Energy expenditure: 12 mo: 31.4±1.4; 24 mo: 29.9±1.5 MET/day Time spent walking: 12 mo: 0.8±0.4; 24 mo: 0.9±0.4 hrs/day Time spent standing: 12 mo: 3.8±1.0; 24 mo: 3.3±0.8 hrs/day Time spent sitting/lying down: 12 mo: 18.5±1.3; 24 mo: 18.1±1.3 hrs/day
Fujita et al. 2013	N: 38 Age: 60.9±9.1 0% male BMI: 23.0±3.6 13.2% bilateral	12 months	Steps: 6163±2410 steps/day LPA: 125±42 min/day MPA: 46±50 min/day VPA: 3±4 min/day
Jeldi et al. 2017	N: 27 Age: 67 (50-82) 33% male BMI: 31 (19-43)	1 year	Steps: 6155±2631 steps/day
Jelsma et al. 2020ab	N: 16 Age: 60 (53-68) 75% male BMI: 29 (20-40) 0% bilateral	10 (8-12) years	Steps: 4600 (1567-11749) steps/day, 50% <5000, 19% ≥8000 steps/day Time cycling: 0.01 (0.0-1.2) hrs/day Time walking: 1.1 (0.4-0.8) hrs/day Time standing: 3.0 (1.6-6.2) hrs/day Time sitting: 9.6 (3.8-13) hrs/day

Arthroplasty/Measurement method/Study	Age/BMI/gender	Follow-up time	Outcome
Kuhn et al. 2013	N: 37 Age: 42.1±7.7 (17.8-50.3) 32% male BMI: 29.0±5.6 (20.1-44.3)	1.3±0.2 years	Steps: 5584±1626 steps/day Inactive time: 71.1±6.1 %/day Low-intensity: 18.7±4.1 %/day Medium-intensity: 8.8%±2.7 %/day High-intensity: 1.4%±0.9 %/day
Matsunaga-Myoji et al. 2020b	N: 107 Age: 61.4±8.1 14% male BMI: 23.1±3.5 0% bilateral	1 & 3 years	Steps 1yr: 6634±3242 steps/day Steps 3yr: 6736±3150 steps/day MVPA 1yr: 58.3±64.6 min/week MVPA 3yr: 72.3±67.4 min/week Meeting guideline of 150min/week MVPA: 18%
Hjorth et al. 2018	N: 71 Age at FU: 68.3 (60.9-69.9) 44% male	8.9 (4.7-10.7) FU at 3, 6, 9, 12 months	% spent on walking, bicycling or high-impact activities during wear time FU + 3mo: 13.43% [12.13-14.72] FU + 6mo: 12.33% [10.50-14.17] FU + 9mo: 10.94% [9.35-12.53] FU + 12mo: 13.34% [11.66-15.03]
Takenaga et al. 2013	N: 55 Age: 39 (18-50) 65% male BMI: 29 (18-42) 5% bilateral	12.1 (10-16) years	1.56 million (77,000-3.26 million) steps/year (4274 steps/day)
Von Rottkay et al. 2018	N: 64 Age: 63.0 53% male BMI: 27.1	12 months	Steps: 5658±2213 steps/day
Wollmerstedt et al. 2010	N: 59 Age: 58 (37-77) 50% male	5 years	Steps: 6564±2795 steps/day
	N: 75 Age: 70 (52-86) 48% male	10 years	Steps: 5101±2412 steps/day
Self-reported questionnaires			
Elman et al. 2014	N: 64 Age: 61.1±13.9 20% male 41% bilateral	2.1 (1-4) years	Questionnaire: YPAS Total time summary index: 23.6±39.1 hrs/week Energy expenditure: 5178.9±6552.4 kcal/week Participation in PA: 189.6±233.1 min/week MVPA: 84.6±158.9 min/week
Jelsma et al. 2020ab	N: 16 THA Age: 60 (53-68) 75% male BMI: 29 (20-40) 0% bilateral	10 (8-12) years	Questionnaire: SQUASH Total activity score: 6150 (1110-18480) MET/week

Arthroplasty/Measurement method/Study	Age/BMI/gender	Follow-up time	Outcome
Ninomiya et al. 2018	N: 58 Age: 68.0±5.4 17.3% male BMI: 22.1±2.8 100% unilateral	10.4±0.4 years	Questionnaire: IPAQ 34.4% highly active (≥1000 kcal/week) 65.5% low-active (<1000 kcal/week)
Paxton et al. 2016	N: 5678 Age: 67 [60-75] 41% male 62% BMI <30 0% bilateral	1-2 years	Questions by nursing staff: Activity: 150 [60-280] min/week Met guidelines: 50%
Rolving et al. 2013	N: 95 Age: 72.3±6.2 33% male	22.4 (18.4-23.9) months	Questionnaire: Physical activity scale PA: 41 [38.5; 48.5] MET/day Leisure activities: 480 [240; 870] min/week
Smith et al. 2018	N: 105 Age: 68.2±9.3 42.9% male BMI: 28.8±4.2	12 months 24 months	Physical activity scale for the elderly (PASE) Total score: 12 mo: 135; 24 mo: 132
Von Rottkay et al. 2018	N: 64 Age: 63.0 53% male BMI: 27.1	12 months	Daily activity questionnaire Load cycles: 4226±948 cycles/day
Wagenmakers et al. 2011	N: 653 Age: 70.3±8.2 26% male BMI: 27.0±4.1	52.4±3.9 weeks	Questionnaire: SQUASH Total activity: 1468.1±1138.3 min/week Leisure activity: 584.5±657.8 min/week LPA: 805.1±800.4 min/week MPA: 333.6±508.0 min/week VPA: 329.4±458.5 min/week 67% met activity guidelines
Wollmerstedt et al. 2010	N: 59 Age: 58 (37-77) 50% male	5 years	Daily activity questionnaire Steps: 6375±2971 steps/day
	N: 75 Age: 70 (52-86) 48% male	10 years	Daily activity questionnaire Steps: 5210±2738 steps/day

Arthroplasty/Measurement method/Study	Age/BMI/gender	Follow-up time	Outcome
Hip resurfacing arthroplasty			
Objective measurement devices			
Jelsma et al. 2020ab	N: 16 Age: 55.5±9.7 (43-67) 75% male BMI: 26.1±3.8 (22-37) 0% bilateral	10 (9-11) years	Steps: 5546 (2274-9966) steps/day, 31% <5000, 13% 8000-10000, 6% >10000 steps/day 10.8±7.0 [4.1-18.3] % time spent active Time cycling: 0.05 (0-0.48) hrs/day Time walking: 1.3 (0.5-1.9) hrs/day Time standing: 3.0 (1.8-5.7) hrs/day Time sitting: 7.6 (4.6-12) hrs/day
Self-reported questionnaires			
Jelsma et al. 2020ab	N: 16 Age: 55.5±9.7 (43-67) 75% male BMI: 26.1±3.8 (22-37) 0% bilateral	10 (9-11) years	Questionnaire: SQUASH Total activity score: 6150 (1110-18480) MET/week
Total knee arthroplasty			
Objective measurement devices			
Bin Sheeha et al. 2020	N: 33 Males: 6 Age: 76±7 (63-85) BMI: 32.38±2.01 Females: 27 Age: 59±6 (49-76) BMI: 37.21±7.65	12 months	Steps: 6174±2287 steps/day Stepping time: 1.36±0.38 hrs/day Time standing: 3.54±0.97 hrs/day Upright time: 4.88±1.47 hrs/day Sedentary time: 19.08±1.54 hrs/day Time <60 steps/min: 593±188 min/week Time 60-100 steps/min: 457±179 min/week Time >100 steps/min (MVPA): 41.7 min/week Met guidelines 150 min MVPA: 24% considering all stepping moments 12% considering >1-min bouts 9% considering >5-min bouts 6% considering >10-min bouts
Brandes et al. 2011	N: 53 Age: 65.8±5.8 34% male BMI: 30.7±4.1	1 year	Gait cycles: 5932±2111 cycles/day Time spent on walking: 12.0±4.7% wear time
Casazza et al. 2020	N: 7 Age: 55.6, SEM 3.5 39% male BMI: 32.8, SEM 1.5	12 months	Steps: 5981 SEM 751 steps/day Energy expenditure: 2297 SEM 133 kcal/day

Arthroplasty/Measurement method/Study	Age/BMI/gender	Follow-up time	Outcome
Daugaard et al. 2018	N: 52 Age: 66±7.2 50% male BMI: 29.5±4.6	5 years	Steps: 6702±3203 steps/day
Hylkema et al. 2020	N: 57 employees working ≥24 hrs/week Age: 59±4 (48-65) 47% male BMI: 20% normal, 33% overweight, 47% obese	1 year	LPA: 36.8% wear time MVPA: 3.1% wear time Sedentary: 60.1% wear time Prolonged sedentary bouts ≥30 min: 16.5% wear time Meeting guidelines 150 min of MVPA/week: 70%
Lutzner et al. 2014	N: 97 Age: 68.9 (CI: 67-70) 54% male BMI: 31.3 (CI: 30.3-32.3)	1 year	Steps: 6473±3654 steps/day MVPA steps: 1935±1728 steps/day MVPA steps: ~19 min/day Walking time: 1.5±0.8 hrs/day 16.5% met PA guidelines
Lutzner et al. 2016	N: 221 Age: 68.1±9.5 43.4% male BMI: 31.3±4.9	1 year	Steps: 6587±3299 steps/day MVPA steps: 1835±1646 steps/day Walking time: 1.6±0.7 hrs/day Patient's lifestyle: 34.8% sedentary, 34.4% low-active, 30.8% active
Matsunaga-Myoji et al. 2020a	N: 58 Age: 72.6±6.0 16% male BMI: 26.1±4.4 58.6% bilateral	2 years	Steps: 4687 (3147-7293) steps/day LPA: 330 (241-469) min/week MVPA: 42.8 (10.0-89.2) min/week Met PA guidelines for older persons (MVPA ≥52.5 min/week): 41.4%
Webber et al. 2017	N: 38 Age: 67.9±7.3 42% male BMI: 30.5±6.1	1 year	Steps: 5935±3316 steps/day Sedentary: 9.2±1.4 hrs/day, 63.8%±10.0 wear time LPA: 291.7±91.0 min/day MVPA: 2.1±10.4 min/day
Wimmer et al. 2015	N: 32 Age: 77.8±6.1 31% male BMI: 29.9±5.7	11.5±3.3 years	Steps: 3102±1553 steps/12hrs Walking: 7.8%±3.8 of time
Self-reported questionnaires			
Groen et al. (2012)	N: 830 Age: 72±9 Gender: 27% male 0% bilateral	(1-6) years	Questionnaire: SQUASH Activity: 1337±1260 min/week 51% met health recommendation 53% met fitness recommendation 46% met both health and fitness recommendation
Hayes et al. 2011	N: 16 Age: 64.4±4.6 50% male BMI: 69.5±2.6	12 months	Energy expenditure: 103 joule/min/kg Activity: 4.6% recording interval

Arthroplasty/Measurement method/Study	Age/BMI/gender	Follow-up time	Outcome
Hodges et al. 2018	N: 434 Age (at FU): 66.0±6.3 47% male	12 months	Questionnaire: Active Australia Survey Met guidelines: 47% ≥6 hrs/day sedentary: 45% Inadequate physical activity and sedentary behavior: 24%
Jones et al. 2012	N: 83 Age: 66±9.7 46% male 0% bilateral	1 year	Historical leisure activity questionnaire (HLAQ): Total activity: 21.4±24.9 MET hrs/week Low-intensity: 0.1±0.9 MET hrs/week Medium-intensity: 19.6±23.6 MET hrs/week High-intensity: 1.5±6.5 MET hrs/week
Kersten et al. 2012	N: 830 Age: 72.0±9.3 25.9% male BMI: 29.4±5.0	3.0±1.2 years	Questionnaire: SQUASH Total activity: 1347±1278 min/week Light PA: 780±874 min/week Moderate PA: 337±577 min/week Vigorous PA: 223±374 min/week Sports activities: 52±140 min/week Walking: 167±335 min/week Cycling: 122±242 min/week 50.8% met physical activity guidelines
Paxton et al. 2016	N: 11084 Age: 68 [62-75] 38% male 49% BMI <30 0% bilateral	1-2 years	Questions by nursing staff: Activity: 150 [60-280] min/week Met guidelines: 50%
Ristolainen et al. 2019	N: 18 sports injury-related TKAs Age: 50.0±5.3 39% male BMI: 27.8±4.1 44% bilateral	4.5±3.0 years	Mailed questionnaire: Leisure time PA: 42.1 MET hrs/week
	N: 45 non-sports injury-related TKA Age: 52.2±5.3 56% male BMI: 29.9±5.5 29% bilateral	6.6±3.7 years	Mailed questionnaire: Leisure-time PA: 18.5 MET hrs/week
Smith et al. 2018	N: 116 Age: 67.3±8.3 41.4% male BMI: 30.1±4.9	12 & 24 months	Physical activity scale for the elderly (PASE) Total score: 12 mo: 121; 24 mo: 142

Arthroplasty/Measurement method/Study	Age/BMI/gender	Follow-up time	Outcome	
Unicompartimental knee arthroplasty − Mixed arthroplasty				
Objective measurement devices				
Blikman et al. (2013) THA/TKA	N: 44 Age: 72±9 23% male BMI: 28±5.4	>1 year	Mean counts/minute: 210±142 Total activity: 99±156 min/week MPA: 92±127 min/week VPA: 7±35 min/week Sitting time: 1252±314 min/week	
Hjorth et al. 2018 THA/HRA	N: 77 Age at FU: 59.1 (51.3-64.4) 66% male	7.1 (4.6-7.6) years and FU + 3, 6, 9, 12 months	% spent on walking, bicycling or high-impact activities during wear time FU + 3m: 12.66% [11.62-13.69] FU + 6m: 11.82% [10.81-12.83] FU + 9m: 11.12% [10.19-12.05] FU+ 12m: 13.15% [12.20-14.09]	
Robertson et al. 2016 THA/HRA/ TKA	N: 13 active patients Age: 48.7 (37-61) 77% male BMI: 23.9 (20.6-28.5) 54% bilateral	8.7 (1.8-15.8) years	Cycles: 8273 (5973-12557) cycles/day % high activity cycles: 39.7% (8-62) % moderate activity cycle: 42.6% (27.1-63.6) % low activity cycle: 18.7% (10.7-30.6) High PA: 4.3%, 58 (11.5-118.1) min/day Moderate PA: 9.4%, 135 (83.5-204.5) min/day	
Self-reported questionnaires			Short-form	Long-form
Blikman et al. (2013) THA/TKA	N: 44 Age: 72±9 23% male BMI: 28±5.4	>1 year	Questionnaire: IPAQ short-form Total activity: 3370±3762 MET min/week Total time: 868±969 min/week MPA: 822±945 min/week VPA: 46±129 min/week Sitting time: 3123±1214 min/week	Questionnaire: IPAQ long-form Total activity: 4498±3840 MET min/week Total time: 1108±922 min/week MPA: 1087±909 min/week VPA: 20±59 min/week Sitting time: 3255±1118 min/week
Poortinga et al. 2014 THA/TKA	N: 658 Age: 68±10.6 32% male BMI: 28.7±4.9	1 year	Questionnaire: SQUASH Total PA: 1525 [1200-1980] min/week Leisure time PA: 420 [210-720] min/week 40% meets PA guideline	

*At time of surgery, unless indicated otherwise.

Abbreviations: BMI: body mass index; FU: follow-up; HPA: high physical activity; HRA: hip-resurfacing arthroplasty; IPAQ: International Physical Activity Questionnaire; LPA: low physical activity; MET: metabolic equivalent of task; MPA: moderate physical activity; MVPA: moderate-to-vigorous physical activity; PA: physical activity; PASE: Physical Activity Scale for the Elderly; SEM: standard error of measurement; SQUASH: Short QUestionnaire to ASsess Health-enhancing physical activity; THA: total hip arthroplasty; TKA: total knee arthroplasty; UKA: unicompartimental knee arthroplasty; VPA: vigorous physical activity; YPAS: Yale Physical Activity Scale